COMPUTATIONAL ARITHMETIC GEOMETRY & SPECTRAL PHYSICS

Spectral Lattice Inversion & Rational $S^3(\mathbb{Q})$ Projection

Algorithmic Architecture, Sequential Function Reference, SU(2) vs. $S^3(\mathbb{Q})$ Comparison, and 100-Prime Verification

100 / 100	23.00%	Z = -0.71	100% Exact
PRIMES SOLVED (P ≤ 541)	RIEMANN EXACT MATCHES	MONTE CARLO (P = 0.7904)	$\mathbb{Q}$ INVARIANTS & GUE CDF

1. Concise Essay on the Algorithm

Overview and Objective. The *Spectral Lattice Inversion* algorithm investigates whether the non-trivial zeros of the Riemann zeta function encode preferential algebraic structure when projected onto four-dimensional quaternion lattices and the rational unit three-sphere $S^3(\mathbb{Q})$. Scaling from an initial 50-prime base case (p_{max} = 229) to 100 prime–zero pairs (p_{max} = 541), the C++17 pipeline couples analytic number theory, O(√p) Diophantine geometry, exact rational field arithmetic, and random matrix spectral verification.

Stage I — Spectral Phase Seeding. For each index $n \in \{1, \ldots, 100\}$, the algorithm pairs the n-th prime p_n with the imaginary ordinate γ_n of the n-th non-trivial Riemann zeta zero $\rho_n = ½ + i\gamma_n$. Evaluating the logarithmic phase $\theta_n = (\gamma_n \ln p_n) \bmod 2\pi$ projects the spectral oscillation onto a target determinant guess $D_{init} = \text{round}((p_n / 2) \cos \theta_n) \in [-\lfloor p_n/2 \rfloor, \lfloor p_n/2 \rfloor]$.

Stage II — O(√p) Coupled Diophantine Lattice Solver. Given (p, D), the solver searches for a Lipschitz integer quaternion $q = (a, b, c, d) \in \mathbb{Z}^4$ satisfying two coupled invariants: the four-square norm $a^2 + b^2 + c^2 + d^2 = p$ and the cross-determinant $ad - bc = D$. Rather than scanning $O(p^2)$ lattice points, the algorithm exploits the algebraic identities $(a \pm d)^2 + (b \mp c)^2 = p \pm 2D$. Using a two-pointer scan to decompose $p + 2D = x_1^2 + y_1^2$ and $p - 2D = x_2^2 + y_2^2$ in O(√p) time, it reconstructs integer coordinates via parity-filtered half-sums. If D_{init} admits no integer solution, an expanding radius search tests $D_{init} \pm k$ (k = 1, 2, …) until the nearest valid lattice shell D_{conv} is reached.

Stage III — Exact Lift to the Rational Field $\mathbb{Q}$. Every converged quaternion $q \in \mathbb{Z}^4$ on $S^3(\sqrt{p})$ is mapped without floating-point approximation into reduced fractions in $\mathbb{Q}$ using Euclidean greatest common divisors. First, the normalized determinant $D_{conv} / p \in \mathbb{Q}$ captures the relative frame orientation. Second, quaternion squaring $U = q^2 / p = ((a^2 - b^2 - c^2 - d^2)/p, 2ab/p, 2ac/p, 2ad/p) \in \mathbb{Q}^4$ embeds every prime onto the true rational unit sphere $S^3(\mathbb{Q})$ with exact unit norm $\|U\|^2 = 1/1$. Third, stereographic projection from the north pole maps (a, b, c, d) onto rational 3-space $(a/(p - d), b/(p - d), c/(p - d)) \in \mathbb{Q}^3$, obeying the exact rational identity $x^2 + y^2 + z^2 = (p - d^2)/(p - d)^2$.

Stage IV — Statistical Benchmarking & GUE Unfolding Calibration. To test statistical significance, a 500-trial Monte Carlo engine replaces Riemann phases with uniform random angles $\theta \sim U(0, 2\pi)$. Across N = 100 primes, Riemann phase projection yields 23 exact matches (23.00%, mean 1.56 steps) versus the random baseline of 25.84 ± 4.00 (25.84%, mean 1.58 steps; Z = -0.71, one-sided p = 0.7904), confirming that zeta-zero phases behave like uniform circular sampling on $S^3(\mathbb{Q})$. Finally, an integrated Gaussian Unitary Ensemble (GUE) solver validates random matrix level repulsion by unfolding 200 × 200 Hermitian eigenvalues via the exact Wigner semicircle cumulative distribution function F(λ), restoring the theoretical ~40.2% peak spacing concentration in $0.75 < s < 1.25$.

2. Architectural Comparison: PDF Complex SU(2) vs. Rational $S^3(\mathbb{Q})$

While the baseline computational physics report projected prime four-square tuples onto complex 2 × 2 special unitary matrices $U(q) \in SU(2)$ over $\mathbb{C}$, our upgraded pipeline projects onto the rational unit three-sphere $S^3(\mathbb{Q})$ via quaternion

squaring and stereographic coordinates. Although $S^3 \cong SU(2)$ are isomorphic as real Lie groups, their arithmetic representations differ fundamentally:

Dimension	Baseline PDF: Complex SU(2) Matrix	Our Pipeline: Rational $S^3(\mathbb{Q})$ Projection
Algebraic Domain	Complex unitary matrices $U(q) \in SU(2) \subset GL(2, \mathbb{C})$ using floating-point arithmetic.	Exact rational field $\mathbb{Q}^4$ and stereographic $\mathbb{Q}^3$ coordinates reduced via `std::gcd`.
Normalization Map	Divides by irrational $\sqrt{p}$: `U(q) = (1/√p) [[a+id, b+ic], [-b+ic, a-id]]`	Squares quaternion in $\mathbb{H}$ and divides by integer p: `U = q² / p = ((a²-b²-c²-d²)/p, 2ab/p, 2ac/p, 2ad/p)`
Precision & Rounding	Introduces IEEE-754 floating-point error when evaluating $\sqrt{p}$ and matrix products (`det U ≈ 1.000`).	Zero rounding error; every coordinate, determinant D/p, and norm $\|U\|^2 = 1/1$ is symbolically exact in $\mathbb{Q}$.
Lattice Search Complexity	$O(p^2)$ brute-force nested loops enumerating all $8(p + 1)$ representations for a single prime p.	$O(\sqrt{p})$ two-square Diophantine solver targeting coupled norm p and cross-determinant ad - bc = D.
Geometric Invariant	Unitarity $U^\dagger U = I$ and special determinant $\det U = (a^2 + b^2 + c^2 + d^2)/p = 1$.	Unit sphere norm $\|U\|^2 = 1$ in $\mathbb{Q}^4$ and stereographic norm $x^2 + y^2 + z^2 = (p - d^2)/(p - d)^2$ in $\mathbb{Q}^3$.
Number-Theoretic Visibility	Obscures residue parity and denominator cancellation inside floating-point decimals.	Exposes exact mod-4 determinant parity, extremal $\pm (p-1)/(2p)$ boundaries, and GCD cancellations.

> **Structural Equivalence Note:** Under the quaternion-to-matrix isomorphism $q = a + bi + cj + dk \mapsto U(q) \in SU(2)$, quaternion squaring $U = q^2/p$ corresponds to matrix squaring $U(q)^2$ in $PSL(2, \mathbb{Q}(i))$, eliminating the quadratic field extension $\mathbb{Q}(\sqrt{p})$ so that every prime shell lands directly inside rational coordinates $\mathbb{Q}^4$.

3. Sequential Function Reference (Execution Order)

The following table lists every function across the six C++ modules in the exact sequential order of program initialization, execution, verification, and output:

#	Function Signature	Source Module	Role & Algorithmic Operation
1	`main(argc, argv)`	`main.cpp`	Top-level entry point orchestrating CLI parsing, zero loading, prime generation, node resolution, verification, CSV export, and Monte Carlo reporting.
2	`parse_cli(argc, argv, cfg)`	`main.cpp`	Parses command-line flags (`--count` , `--max-steps` , `--mc-trials` , `--seed` , `--verify-gue` , `--output`) into `RunConfig` .
3	`print_usage(prog)`	`main.cpp`	Displays formatted CLI option documentation when `--help` or an invalid flag is encountered.
4	`load_gammas_from_file(...)`	`io_and_montecarlo.hpp`	Optional parser for external LMFDB/ Odlyzko zero tables; skipped when using the embedded 100-zero table `DEFAULT_RIEMANN_GAMMAS` .
5	`generate_primes(n)`	`lattice_solver.hpp`	Executes a dynamically bounded Sieve of Eratosthenes using Rosser–Schoenfeld upper bounds to return the first n primes ($p_1 = 2 \dots p_{100} = 541$).
6	`resolve_from_initial_d(...)`	`lattice_solver.hpp`	Tests $D_{initial}$ for an exact lattice match; if none exists, searches outward by ±1, ±2, … up to `max_steps` to find $D_{converged}$.
7	`solve_lattice(p, D, out)`	`lattice_solver.hpp`	Solves $a^2+b^2+c^2+d^2 = p$ and $ad - bc = D$ in $O(\sqrt{p})$ time by pairing signed two-square solutions of $p + 2D$ and $p - 2D$ with parity checks.
8	`solve_two_squares_all(target)`	`lattice_solver.hpp`	Two-pointer scan finding all non-negative integer pairs (X, Y) with $X \geq Y$ such that $X^2 + Y^2 =$ target.
9	`Rational::Rational(n, d)`	`spectral_inversion.h`	Constructs canonical reduced fractions in $\mathbb{Q}$ by normalizing denominator sign and dividing numerator and denominator by `std::gcd(|n|, d)` .
10	`lift_to_rationals(p, d, q)`	`rational_projection.hpp`	Computes exact rational determinant D/p, unit quaternion squared norm $\|q^2/p\|^2 = 1/1$ on $S^3(\mathbb{Q})$, and stereographic coordinates in $\mathbb{Q}^3$.
11	`verify_node_invariants(n, lbl)`	`main.cpp`	Validates $\mathbb{Z}^4$ norm (p) and determinant (D_{conv}) invariants on every Riemann and baseline node, aborting with exit code 2 on any mismatch.
12	`verify_rational_invariants(n)`	`rational_projection.hpp`	Symbolically verifies unit sphere norm `1/1` and stereographic identity $a^2 + b^2 +$

#	Function Signature	Source Module	Role & Algorithmic Operation
			$c^2 = p - d^2$ in $\mathbb{Q}$ without floating-point rounding.
13	`export_side_by_side_csv(...)`	`io_and_montecarlo.hpp`	Writes all 100 paired Riemann and baseline rows—including $\mathbb{Z}^4$ tuples and $\mathbb{Q}$ fractions—to `spectral_comparison_100.csv`.
14	`Rational::str() / operator<<`	`spectral_inversion.h`	Formats reduced rational numbers cleanly as integers (when `den == 1`) or `num/den` strings for console and CSV output.
15	`run_monte_carlo_comparison(...)`	`io_and_montecarlo.hpp`	Runs 500 seeded random-phase trials across all 100 primes to compute baseline mean exact matches, standard deviation, Z-score, and empirical p-value.
16	`verify_gue_semicircle_unfolding(...)`	`gue_unfolding.hpp`	Generates a 200 × 200 GUE complex Hermitian matrix, unfolds bulk eigenvalues via the Wigner semicircle CDF, and verifies ~40% peak spacing density.
17	`compute_hermitian_eigenvalues(H)`	`gue_unfolding.hpp`	Diagonalizes complex Hermitian matrix H to machine precision via Householder tridiagonalization followed by implicit QL shifts.

4. Key Empirical Findings in $\mathbb{Q}$ (N = 100 Scaling)

- **Mod-4 Determinant Parity Law:** Across all 99 odd primes, D_{conv} is 100% even whenever $p \equiv 1 \pmod 4$ (47/47 cases) and 100% odd whenever $p \equiv 3 \pmod 4$ (52/52 cases), forced by $p + 2D \equiv 1 \pmod 4$.
- **Extremal Rational Boundary Saturation:** Exactly 15% of Riemann nodes saturate the theoretical limit $D_{conv} / p = \pm(p - 1)/(2p)$, collapsing one two-square factor to $1^2 + 0^2 = 1$ and yielding compact stereographic coordinates (e.g., Index 42, $p = 181 \rightarrow (1/19, 0, 0)$).
- **Orthogonal Splitting Exclusivity:** Zero determinant ($D_{conv} = 0$, $ad = bc$) occurs exclusively on Pythagorean primes $p \equiv 1 \pmod 4$ via Fermat's Two-Square Theorem (e.g., Index 45, $p = 197 = 14^2 + 1^2$).
- **Stereographic Radial Shell Scaling:** Across all 100 primes, the squared stereographic norm $x^2 + y^2 + z^2 = (p - d^2)/(p - d)^2$ in $\mathbb{Q}^3$ scales tightly with $1/p$ (mean ratio 0.858 ± 0.160, median 0.917).